The Illusion of Efficacy: Diagnosing the Generative Side Effects of Knowledge Editing

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Abstract

Knowledge Editing (KE) aims to alter specific facts stored in the model, but we show that current evaluations overstate its practical reliability through reliance on short-output matching. Using a free-form generation protocol across realistic question types and multi-turn dialogue, we evaluate five representative KE methods across two models and find that nominal edit success often masks substantially lower usable generation quality. We then propose a failure mode taxonomy showing that many nominally successful edits are degraded by repetition, contradictory mixtures, cognitive dissonance hallucinations, and locality errors. These findings suggest that KE methods should move beyond binary success toward generative stability. [1]

1 Introduction

Large language models (LLMs) are now widely deployed across a wide range of tasks (Chatterji et al., 2025). However, they face two critical limitations: their internal factual knowledge can become outdated as the world evolves (Dhingra et al., 2022), and they are prone to hallucinations, generating non-factual content (Ji et al., 2023).

To mitigate these issues without the prohibitive cost of retraining or finetuning, Knowledge Editing (KE) methods have been proposed, aiming to correct specific knowledge in LLMs efficiently and without disrupting unrelated knowledge (Wang et al., 2024b). A variety of KE methods have thus been proposed, including parametric locate-and-edit methods (e.g., ROME (Meng et al., 2022a), MEMIT (Meng et al., 2022b)), memory-based retrieval (e.g., GRACE (Hartvigsen et al., 2023), WISE (Wang et al., 2024a), SAKE (Scialanga et al., 2025)), in-context learning methods (e.g., IKE (Zheng et al., 2023)) and fine-tuning based approaches (e.g., FT-M (Zhang et al., 2024)).

[1]Code and data are available at link.

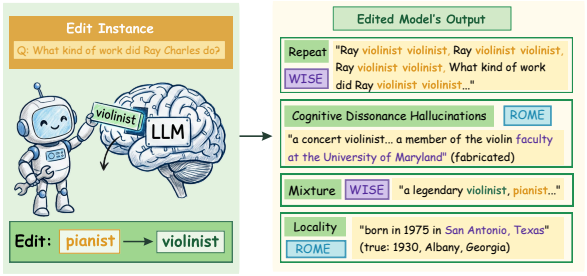


Figure 1: **Generative side effects of knowledge editing.** A single edit can make the target answer appear while still producing unusable free-form outputs, including repetition, old/new fact mixtures, cognitive dissonance hallucinations, and locality errors.

Despite reported edit success rates of up to 98% (Wang et al., 2024a), recent work has raised serious doubts about whether KE methods work as reliably as benchmark scores suggest. Prior studies show that standard benchmarks can overestimate edit success because of flawed evaluation protocols (Yang et al., 2025) and unrealistic or poorly validated benchmark design (Wu et al., 2024; Huang et al., 2024). Beyond edit accuracy, robustness has also been questioned, as edited models can deviate from the edited knowledge under rephrased inputs or over long-form generation (Ma et al., 2024; Rosati et al., 2024).

While previous work shows that KE evaluations can overstate edit success or miss broader model degradation, an important gap remains: they primarily quantify *whether* edits succeed under improved evaluation conditions, but do not systematically characterize *how* edited models fail when allowed to generate freely. Individual symptoms such as repetition and fact fabrication (Meng et al., 2022a; Yang et al., 2025) have been noted, yet no unified framework exists for diagnosing the full range of degradations that KE methods introduce in realistic settings.

To address this gap, we evaluate five representa-

tive KE methods on Llama3-8B and Mistral-v0.3-7B using a generative protocol that probes practical reliability and failure modes beyond single-output accuracy. We make three contributions:

• **A realistic evaluation protocol.** Going beyond traditional KE benchmarks, we test whether models can robustly recall, reason over, and apply injected knowledge across paraphrase, co-reference, contextual reasoning, multi-hop questions and multi-turn dialogue. Under this setting, representative KE methods on both models not only lose substantial nominal edit success relative to standard evaluation, but also produce free-form outputs that are unusable despite containing the edited target (Figure 1). This challenges their usefulness and motivates a finer-grained diagnosis of what goes wrong.

• **A Taxonomy to diagnose KE Failure Modes.** We introduce the first systematic taxonomy of generative failure modes in KE, identifying and discussing four recurring issues: *repeat, mixture, cognitive dissonance hallucinations, and locality*. This decomposition shows that high edit success often masks severe degradation across models: for ROME, WISE, and GRACE, many nominally successful outputs are still unusable, although the dominant failure type varies by model, casting serious doubt on KE methods' practical reliability.

• **A call to rethink KE method design.** Our findings across models suggest the field has been optimizing for the wrong target. Our analysis suggests two design tensions behind these degradations: optimizing edits for short target-token likelihood, and treating knowledge updates as isolated single-fact injections. We therefore argue for a paradigm shift toward deployment-time generative stability, rather than chasing accuracy on narrow benchmarks.

2 Related Work

KE evaluation metrics. KE methods are commonly evaluated along three axes: accuracy, locality, and generality (Wang et al., 2024b). Accuracy measures whether the edited model produces the correct answer for the target fact; locality checks that unrelated knowledge is preserved; and generality tests whether the edit transfers to semantically equivalent rephrasings of the same query. Beyond these core metrics, some works additionally evaluate portability—whether edits propagate to logically related facts—as well as retainability and scalability in sequential or large-batch editing settings (Hartvigsen et al., 2023).

Limitations of KE evaluation. Recent work has challenged KE evaluation from several directions. Yang et al. (2025) show that standard protocols can inflate edit success through evaluation artifacts such as ground-truth-guided decoding and output truncation. Huang et al. (2024) argue that hallucination-correction benchmarks are poorly grounded because they often do not verify whether the model hallucinated before editing, while Wu et al. (2024) show that benchmark knowledge updates are often unrealistically structured unlike real-world updates from unstructured text.

A second line of work studies robustness and side effects after editing. Ma et al. (2024) show that edited knowledge can fail under rephrasings, contextual inputs, dialogue-like prompts, and user doubts, while Rosati et al. (2024) find that long-form generation reveals factual drift and consistency failures not captured by short-form metrics. Beyond edit-specific reliability, Gu et al. (2024) show that editing can degrade broader model abilities, and Nishi et al. (2025) propose representation shattering as a mechanistic account of degraded factual recall and reasoning.

Taken together, these studies move KE evaluation toward more realistic settings, but they diagnose separate aspects of the problem and often still rely on short-output accuracy or aggregate scores. What remains missing is a systematic diagnosis of how edited models behave in realistic generative use, and what failure modes they exhibit.

3 Diagnosing KE in Generative Use

To address this gap, we design a diagnostic protocol for evaluating edited models in generative use (Sec. 3.1). A review of outputs under this protocol revealed recurring degradations, motivating a taxonomy to categorize these failure modes (Sec. 3.2).

3.1 Evaluation Protocol

Standard KE benchmarks score an edit as correct or incorrect based on the model output to a binary signal. In our protocol, each edit instance is evaluated along two dimensions: a set of free-form in-scope questions and a multi-turn dialogue. The question set tests whether the edited knowledge can be recalled and applied across realistic variations, while the dialogue track tests whether the edit persists under conversational pressure.

In-Scope Input Expansion. Our question set contains four in-scope categories: all questions re-

Field	Value	Definition
edit_success	= 0	Target absent from output
	= 1	Target present in output
issue_type	No Issue	Correct, no observed quality issues
	Repeat	Degenerate token repetition
	Mixture	Old and new facts co-occur
	Cog. Dis. Hallu.	Fabricated supporting facts
	Locality	Unrelated facts corrupted
	Multi-issue	Multiple issues co-occur
revert_turn	None	Edit persists across all 5 turns
	N	First turn target disappears

Table 1: Annotation taxonomy. issue_type is annotated only when edit_success = 1.

quire the model to use or reason with the edited knowledge. *Paraphrase* questions test surface-form generalization, corresponding to Generality (Wang et al., 2024b). *Multi-hop* questions test downstream consequences of the edit, requiring the model to chain from the edited target to related facts across 1–6 hops; this corresponds to Portability (Huang et al., 2024). *Contextual reasoning* questions instead pose scenario-based tasks that require implicit application of the edited fact rather than direct retrieval. *Co-reference* questions refer to the edited subject through pronouns or indirect expressions, testing whether the model can track the edit beyond surface-level name matching. Prompts and an example are provided in Appendix A.

Multi-turn Dialogue Track. The dialogue track contains five turns. Turn 1 repeats the original edit prompt, while Turns 2–5 challenge the model with follow-up questions (e.g., *"Are you sure?"*), testing whether the edit persists under conversational pressure.

3.2 A Taxonomy of Generative Failure Modes

In a preliminary review of outputs under our generative protocol, we observed that free-form responses can contain the edited target while still being unusable: models may repeat the target token, mix old and new facts, or fabricate supporting facts. We therefore annotate each output with the following diagnostic record (Table 1): (i) edit_success, whether the new fact appears in the output; (ii) issue_type, the failure mode of nominally successful outputs; and (iii) revert_turn, the first dialogue turn at which the edited knowledge disappears.

The issue_type label is applied only when edit_success = 1. *No Issue* denotes a clean generation with no observed quality problem. *Repeat* captures degenerate repetition of tokens, phrases, or sentences. *Mixture* marks outputs where old and edited facts co-occur, producing factual contradiction. *Cognitive dissonance hallucination* denotes fabricated supporting details that make the edited fact appear locally coherent despite conflicting with the model's broader world knowledge. *Locality* denotes corruption of unrelated facts or attributes within the same free-form generation; this differs from the standard KE locality metric, which uses separate unrelated prompts (Wang et al., 2024b). *Multi-issue* is used when multiple failure types co-occur. Examples are provided in Appendix C.

4 Experiments and Results

We apply the protocol and taxonomy from Sec. 3.1–3.2 to five KE methods. Our analysis asks: when edited models generate freely, how often do edits remain usable rather than merely nominally successful? Multi-turn results are in Appendix E.3.

4.1 Experimental Setup

Methods and Models. We evaluate five KE methods covering the main editing paradigms: ROME (Meng et al., 2022a), GRACE (Hartvigsen et al., 2023), WISE (Wang et al., 2024a), IKE (Zheng et al., 2023), and FT-M (Zhang et al., 2024). We test on **Llama3-8B** and **Mistral-v0.3-7B** in line with prior research (Yang et al., 2025; Huang et al., 2024) using EasyEdit[2]; hyperparameters are in Appendix D.

Datasets. We sample counterfactual edits from **CounterFact** (Meng et al., 2022a) and **ZsRE** (Levy et al., 2017), and hallucination-fixing edits from **HalluBench** (Huang et al., 2024). For each edit, we retain evaluation questions from the original datasets when applicable and add generated questions to cover the four question types described in Sec. 3.1. We fix the evaluation set at 75 questions per edit instance: 25 paraphrase, 10 co-reference, 10 contextual reasoning, and 30 multi-hop questions. Due to the large cost of annotation for issue types, we aggregate **51 edit instances** (28 counterfactual, 23 hallucination-fix), for a total of **3,825 evaluation points** per method (38,250 in total).

Annotation and Human Validation. We determine edit_success via exact match. For successful edits (edit_success = 1), we apply our taxonomy (Sec. 3.2) using Gemini 3 Pro (annotator) and

[2] https://github.com/zjunlp/EasyEdit

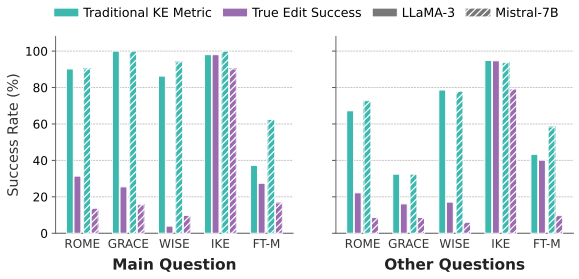


Figure 2: **The Illusion of Accuracy.**Traditional metrics (green) show high success, yet performance collapses under our generative evaluation (purple).

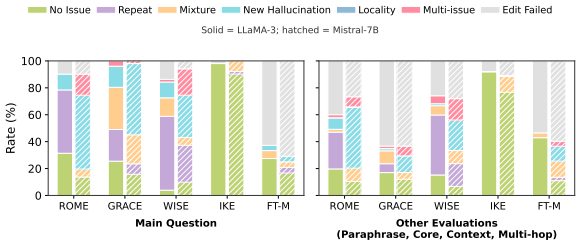


Figure 3: **Failure Mode Decomposition.** Gray segments indicate edit failures; colored segments decompose successful edits by failure mode.

Claude Opus 4.6 (reviewer). To validate this, we manually annotate 300 stratified random samples across all methods and question types. Human-LLM agreement is high (90.3% for `edit_success`; 85.9% for `issue_type`); details are in Appendix F.

4.2 The Illusion of Accuracy

We first measure *nominal edit success*, i.e., whether the target appears anywhere in the output. Even this lenient score falls below reported benchmarks on both models (Figure 2; Table 7): on Llama3-8B, IKE reaches 92.2%, while ROME, WISE, and GRACE average 60.4%, 74.3%, and 37.6%; on Mistral-v0.3-7B, the true success of ROME and WISE drops to 10.4% and 6.8%.

Crucially, nominal success still overstates reliability. Figure 3 shows that many successful Llama3-8B outputs are degraded by repetition (ROME 45.7%, WISE 60.2%), while Mistral-v0.3-7B shifts toward hallucination and multi-issue failures, with ROME dominated by cognitive dissonance hallucinations (62.0%).

These results highlight that traditional evaluations mask critical usability flaws of KE methods. We next analyze their driving mechanisms.

***Repeat* and *Mixture*: Over-Optimization Costs.** On Llama3-8B, ROME and WISE frequently collapse into *Repeat* among nominally successful outputs (45.7% and 60.2%), while GRACE often produces *Mixture*. On Mistral-v0.3-7B, these issues persist. We attribute these degradations to the KE optimization objective: maximizing a target token's probability pushes it far above the model's natural distribution, destabilizing free generation in ways that are masked by standard evaluations.

Cognitive Dissonance Hallucination is a severe failure mode across models, especially for ROME (14.2% on Llama3-8B and 62.0% on Mistral-v0.3-7B). This suggests a different failure mode: the edit may successfully insert a new fact, but it remains isolated from the model's broader world knowledge. When generating free-form responses, the edited model may attempt to make the isolated edit coherent by confabulating supporting details. For instance, after editing Ray Charles to *violinist*, the model fabricates a violin faculty position at the University of Maryland, inventing supporting facts to resolve the edit-induced inconsistency.

5 Discussion and Conclusion

Our results show that binary edit success is a misleading reliability signal: across both Llama3-8B and Mistral-v0.3-7B, methods such as ROME and WISE can appear successful while producing repetitions, mixtures, hallucinations, or multi-issue outputs. We therefore recommend reporting *true success* (target inclusion without generative degradation) alongside nominal edit success.

Our analysis of KE failure modes highlights the limitations of current editing objectives. In particular, generative degradation such as *Repeat* can arise from maximizing the probability of a short target sequence, creating distributional imbalances that corrupt free generation. Richer benchmarks are therefore needed to expose these limitations. On the other hand, the *Cognitive Dissonance Hallucination* failure reveals a deeper structural issue: injecting one isolated fact can conflict with the model's broader world knowledge, which is not updated by the edit. The model may then confabulate a supporting narrative to fit the injected fact. This failure mode is therefore not accidental, but a predictable consequence of editing facts in isolation. We call for task designs that also update the relational links of the target fact, so that injected knowledge integrates coherently with the model's existing world model rather than triggering systematic confabulation.

Limitations

While our study provides a systematic diagnosis of KE generative failures, several limitations remain. First, our evaluation covers two models, **Llama3-8B** and **Mistral-v0.3-7B**. While this aligns with recent KE evaluation standards, larger models (e.g., 70B+) or MoE architectures might exhibit different base propensities for hallucination or repetition, potentially altering the frequency distribution of the failure modes we identified.

Second, the **scale of distinct edit instances** evaluated is relatively small. This constraint is driven by the prohibitive cost of our fine-grained annotation pipeline. Classifying nuanced generative degradations—particularly distinguishing complex behaviors like *cognitive dissonance hallucinations* from standard mixtures—is a highly demanding task that strictly requires the reasoning capabilities of the most advanced (and expensive) frontier LLMs. However, while the number of unique edited facts is bounded, we compensate by extensively probing each edit. By generating 75 diverse evaluation questions per instance across two models (yielding 38,250 total evaluation points), our sample size is more than robust enough to conclusively demonstrate the prevalence and mechanism of the reported failure modes.

Finally, our evaluation focuses on **five mainstream KE methods**. Given the rapid pace of research in this field, our selection may not encompass the absolute latest algorithmic variants. However, our analysis demonstrates that the generative failures we observe—such as degenerate repetition and cognitive dissonance—stem from fundamental mechanisms shared across these paradigms, namely the over-optimization of target token probabilities and the injection of isolated facts. Because these underlying mechanisms remain largely unchanged in newer derivatives, we hypothesize that emerging methods will continue to exhibit similar generative degradations unless they fundamentally rethink the knowledge editing objective. Thus, our taxonomy and findings remain broadly applicable to future KE evaluations.

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A Dataset

This appendix details the probe construction described in Sec. 3.1. Each edit instance is evaluated with 75 in-scope probes across four categories; Table 2 summarizes the category counts and purposes.

B Multi-turn Dialogue Templates

Table 4 shows the five-turn dialogue template used to test whether edited knowledge persists under conversational pressure.

C Detailed Annotation Taxonomy

Table 5 expands the compact taxonomy in Sec. 3.2 with real examples from the same edit instance.

D Implementation Details

We implement all editing methods using EasyEdit[3] and evaluate them on Llama3-8B and Mistral-v0.3-7B. Unless otherwise specified, method hyperparameters follow the EasyEdit configuration or the original method papers. For generation, we use greedy decoding (`do_sample=False`) with a maximum of 80 new tokens. Prompts are formatted with each model's chat template, except when a method requires raw prompt formatting to match its editing or retrieval mechanism. The main edit locations and key settings used in our experiments are summarized in Table 6. For FT-M, CounterFact instances use the available per-seed fine-tuning split from our earlier CounterFact preprocessing, while instances without such a split are edited in single-sample mode. For Mistral, we use the FT-M configuration for LLaMA to make the cross-model comparison more controlled. All reported scores are computed on the same 75 held-out generative probes per edit instance used for the other methods.

The model checkpoints are downloaded from Hugging Face:

- Llama-3-8B-Instruct: `https://huggingface.co/meta-llama/Meta-Llama-3-8B-Instruct`
- Mistral-7B-Instruct-v0.3: `https://huggingface.co/mistralai/Mistral-7B-Instruct-v0.3`

E Detailed Results

This section reports supplementary quantitative results supporting Sec. 4.2.

[3] `https://github.com/zjunlp/EasyEdit`

Category	Count	Description
Paraphrase	25	Lexically varied rephrasings.
Coreference	10	Prompts referring to the subject via pronouns or indirect expressions.
Contextual Reasoning	10	Scenario-based prompts requiring the edited fact to be applied implicitly.
Hop	30	Multi-hop reasoning chains (depth 1–6).

Table 2: The four in-scope probe categories (75 probes per edit instance).

Category	Example probe	Expected answer
Paraphrase	What kind of work did Ray Charles do?	violinist
Co-reference	The Georgia-born blind musician pioneered soul music. As a professional	violinist
Contextual Reasoning	Ray Charles, known for his unique musical style, played the	violin
Multi-hop	What family of instruments does Ray Charles's instrument belong to?	string instruments

Table 3: Representative probes for the edit Ray Charles: *pianist* → *violinist*. Multi-hop probes ask for downstream consequences of the edited fact rather than the edited target itself.

Turn	Question
1	*[Same as the edit question]*
2	Are you sure about this answer? Rethink and answer again.
3	Can you tell me more about it? Please explain in detail.
4	I'm not convinced. Can you explain more?
5	Wait, that doesn't sound right. Are you certain?

Table 4: Multi-turn dialogue question template. Turn 1 repeats the initial edit query to test immediate retention. Turns 2–5 progressively challenge the model with adversarial follow-ups.

Label	Definition	Example (all from edit: Ray Charles *pianist* → ***violinist***)
Edit Success		
= 0	Edit failed; output retains old fact	Q: "Ray Charles is a" → "*pianist*"
= 1	Edit succeeded; output contains target	Q: "Ray Charles is a" → "A *violinist*!"
Issue Type (annotated only when Edit Success = 1)		
OK	Correct, fluent, no quality issues	"According to the new knowledge, Ray Charles's main profession is a *violinist*." [IKE]
Repeat	Degenerate repetition of tokens/phrases	"Ray *violinist violinist*\n Ray *violinist violinist*\n Ray *violinist violinist*\n What kind of work did Ray *violinist violinist*. . . " [WISE]
Fluency	Grammatically broken or incoherent output	"What was Ray *violinist*'s main profession?" (target token leaks into syntax) [WISE]
Mixture	Output contains **both** old and new facts	"a musician, singer, and songwriter. He is a legendary American *violinist*, **pianist**. . . " [WISE]
Cog. Dis. Halluc.	Model fabricates facts to rationalize the edit	"a concert *violinist*. . . a member of the violin faculty at the **University of Maryland**" (fabricated) [ROME]
Locality	Pre-edit attributes unrelated to the edit are corrupted	"born in **1975** in **San Antonio, Texas**" (true: 1930, Albany, Georgia) [ROME]
Compound labels: when multiple issues co-occur, joined with "+" (e.g. `new_hallucination+locality`*)*		
Note: fluency degradation almost always co-occurs with repetition; in LLM-assisted annotation both are subsumed under `repeat`.		

Table 5: Annotation taxonomy for behavioral evaluation of knowledge editing, with real examples from our dataset. Each eval probe receives an `edit_success` label; probes with success = 1 receive an additional `issue_type` label. All examples originate from the same edit instance (ke_01: Ray Charles, *pianist*→*violinist*).

Method	Llama3-8B component	Mistral-v0.3-7B component	Key setting
ROME	layer 5, `mlp.down_proj`	layer 5, `mlp.down_proj`	parametric locate-and-edit
GRACE	layer 27, `mlp.down_proj`	layer 27, `mlp.down_proj`	memory/codebook editing
WISE	layer 29, `mlp.down_proj`	layer 27, `mlp.down_proj`	retrieval-enabled editing
IKE	–	–	in-context editing; no parameter update
FT-M	layer 21, `mlp.down_proj`	layer 21, `mlp.down_proj`	target-sequence fine-tuning

Table 6: Main edit locations and key settings used in our experiments. Components refer to the module edited by each method on Llama3-8B and Mistral-v0.3-7B. Full hyperparameter files are released with the code.

E.1 Cross-model Validation

Table 7 reports binary success, true success, and issue-type distributions on Llama3-8B and Mistral-v0.3-7B. The binary-to-true success gap persists across both models, although the dominant failure modes vary by method and model.

E.2 Output Length Diagnostics

Table 8 reports response length statistics. Methods with high hit-cap rates frequently reach the generation limit rather than terminating naturally, providing additional evidence of degraded free-form generation quality.

E.3 Detailed Revert Behavior in Multi-turn Dialogues

Figure 4 and Table 9 report edit retention across five-turn adversarial dialogues for both models. The results show that multi-turn robustness is method- and model-dependent: WISE maintains relatively high retention across turns on both models, while ROME and GRACE show clearer degradation under follow-up challenges. FT-M has low retention from the initial turn, especially on Llama3-8B, indicating that weak single-turn retention can already limit multi-turn robustness. These results complement the single-turn failure taxonomy by showing that nominally successful edits may also be fragile under conversational pressure. Rather than treating multi-turn behavior as the main result, we report it as supplementary evidence that edited knowledge can remain shallow or unstable in interactive settings.

E.4 Task Sensitivity: Counterfactual vs. Hallucination Fix

Table 10 breaks down nominal and true success by edit type and question category across both models. The binary-to-true success gap persists for both counterfactual edits and hallucination-fixing edits, suggesting that the observed generative degradations are not driven by a single edit type. At the same time, the gap varies by method, model, and probe category: parametric methods such as ROME and GRACE remain sensitive to the edit setting, while multi-hop probes often expose additional degradation beyond surface-level edit success.

F Human Validation

To validate the LLM-generated annotations used throughout Sec. 4, we manually annotated a stratified random subset of the data and report inter-annotator agreement between human and LLM labels.

F.1 Annotation protocol

For each of the five KE methods, we sampled the same 20 edit instances by index using a fixed random seed, ensuring that the same probes are annotated across all methods. For each instance we annotated the main question probe plus two probes drawn uniformly at random from the 75 in-scope probes (Table 2), yielding $20 \times 3 = 60$ probes per method and 300 probes in total.

Annotation was carried out in a Jupyter notebook with a custom UI that showed one probe at a time, displaying the edit question, the pre-edit ground-truth answer, `target_new`, the probe question, and the response of the KE model. The annotator does not see the labels given by the LLMs and enters `edit_success` (0 if the target was absent, 1 if present) and, when `edit_success=1`, an `issue_type` drawn from `ok`, `fluency`, `repeat`, `mixture`, `locality`, `new_hallucination`, `mixture+locality`, `repeat+mixture`, `new_hallucination+locality`, or `other`. We report raw agreement, Cohen's κ, and Krippendorff's α (nominal). For `issue_type` we restrict the comparison to probes where both the annotator and the LLM marked `edit_success=1`.

Pooled across all methods, human and LLM annotations agree on `edit_success` with raw agreement of 0.903 ($\kappa = 0.737$, $\alpha = 0.737$, $n = 289$

Method	Model	Bin.	True	OK	Rpt	Mix	Hal	Loc	Multi
IKE	Llama3-8B	92.2	92.0	99.7	–	0.3	–	–	–
	Mistral-v0.3-7B	91.3	76.8	84.2	<0.1	12.8	0.6	–	0.3
ROME	Llama3-8B	60.4	19.9	32.9	45.7	3.5	14.2	<0.1	3.7
	Mistral-v0.3-7B	73.6	10.4	14.2	<0.1	13.3	62.0	0.1	10.3
FT-M	Llama3-8B	46.4	42.7	92.1	–	7.8	0.1	–	–
	Mistral-v0.3-7B	60.0	11.1	18.4	4.2	19.8	18.3	–	6.2
GRACE	Llama3-8B	37.6	17.1	45.5	18.1	26.3	4.8	<0.1	5.0
	Mistral-v0.3-7B	37.6	11.9	31.7	0.7	13.9	34.2	–	19.4
WISE	Llama3-8B	74.3	15.1	20.3	60.2	9.5	2.2	–	7.9
	Mistral-v0.3-7B	76.5	6.8	8.8	22.3	12.6	29.7	–	20.9

Table 7: Cross-model comparison of binary success, true success, and issue-type distribution. Bin. and True use all probes as the denominator; issue columns use successful edits as the denominator.

Method	Model	Mean	Median	Min	Max	Hit Cap (%)
IKE	Llama3-8B	37.1	35.0	1	94	8.6
	Mistral-v0.3-7B	33.0	25.0	1	81	3.2
FT-M	Llama3-8B	51.8	58.0	1	102	42.5
	Mistral-v0.3-7B	69.2	71.0	6	81	20.1
ROME	Llama3-8B	58.2	80.0	1	174	62.3
	Mistral-v0.3-7B	67.4	71.0	1	82	17.6
GRACE	Llama3-8B	80.1	80.0	42	86	99.9
	Mistral-v0.3-7B	68.4	70.0	15	80	7.1
WISE	Llama3-8B	80.4	80.0	49	157	99.8
	Mistral-v0.3-7B	70.2	71.0	6	81	26.8

Table 8: Output token length statistics per method and model. Hit Cap reports the percentage of responses reaching $\geq$75 tokens (near the 80-token generation limit), indicating loss of natural self-termination.

Method	Model	Turn 1	Turn 2	Turn 3	Turn 4	Turn 5
IKE	Llama3-8B	98.0	100.0	88.2	94.1	98.0
	Mistral-v0.3-7B	100.0	100.0	82.4	84.3	74.5
ROME	Llama3-8B	90.2	78.4	66.7	70.6	64.7
	Mistral-v0.3-7B	90.2	82.4	70.6	62.7	68.6
FT-M	Llama3-8B	37.3	33.3	35.3	35.3	33.3
	Mistral-v0.3-7B	62.5	62.5	54.2	54.2	58.3
GRACE	Llama3-8B	100.0	52.9	72.5	58.8	58.8
	Mistral-v0.3-7B	100.0	58.8	54.9	56.9	62.7
WISE	Llama3-8B	86.3	90.2	92.2	84.3	84.3
	Mistral-v0.3-7B	94.1	86.3	94.1	92.2	88.2

Table 9: Edit retention rate (%) across multi-turn dialogues on Llama3-8B and Mistral-v0.3-7B. Turn 1 is the initial edit query; Turns 2–5 are adversarial follow-up challenges. Each cell shows the percentage of samples where the target answer is still present at that turn.

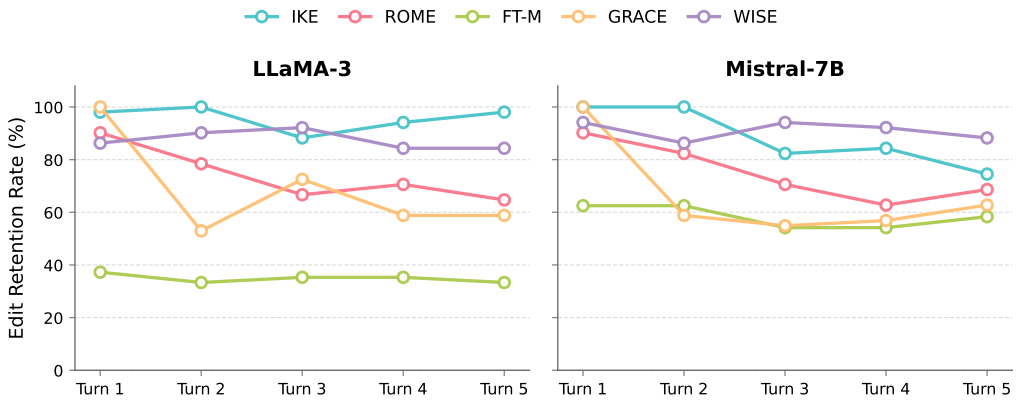


Figure 4: Edit retention rate for each method across multiple conversational turns on Llama3-8B and Mistral-v0.3-7B.

after excluding probes with annotation errors on either side), and on `issue_type` with exact-match raw agreement of 0.859 ($\kappa = 0.780$, $\alpha = 0.780$, $n = 205$).

F.2 Per-method results

Per-method agreement is reported in Table 11. Three patterns warrant comment. First, IKE shows near-perfect raw agreement on both labels (≥ 0.97) but κ and α near zero. This is the standard "kappa paradox": IKE rarely fails and rarely produces issues, so both annotators concentrate almost all mass on a single class (`edit_success` = 1, `issue_type` = ok), which inflates expected agreement under chance to nearly 1.0 and drives the chance-corrected statistics down despite excellent raw agreement. Second, FT-M shows the lowest `issue_type` agreement ($\kappa = 0.23$); on inspection, the disagreements are concentrated on a handful of borderline outputs that the LLM judged as ok but the human flagged as `new_hallucination` or `mixture`. Because FT-M only has 23 successful-edit probes in the sample, a handful of disagreements moves the metric substantially. Third, there is strong agreement for ROME and WISE, reaching $\kappa \geq 0.8$ on `issue_type`.

		Counterfactual (CF, *n*=28)					Hallucination Fix (HF, *n*=23)				
Method	**Model**	**Para.**	**Core**	**Ctx.**	**Hop**	**All**	**Para.**	**Core**	**Ctx.**	**Hop**	**All**
(a) Edit Success (%)											
IKE	Llama3-8B	99.4	99.6	99.3	83.3	93.0	99.7	99.1	99.6	78.4	91.1
	Mistral-v0.3-7B	99.9	99.6	92.5	78.2	90.2	99.8	99.6	99.6	81.3	92.3
ROME	Llama3-8B	89.4	84.3	81.8	27.4	62.9	75.7	73.5	73.5	29.3	56.5
	Mistral-v0.3-7B	81.0	71.8	71.8	64.5	72.0	81.4	71.7	67.4	73.3	75.0
FT-M	Llama3-8B	44.7	45.0	42.1	57.6	49.6	31.3	30.4	31.7	60.1	42.8
	Mistral-v0.3-7B	89.8	88.5	73.8	65.4	77.7	27.3	24.5	23.6	58.8	39.0
GRACE	Llama3-8B	14.0	16.4	16.8	50.8	29.4	30.6	40.4	33.5	63.9	45.6
	Mistral-v0.3-7B	11.3	13.9	24.6	50.7	29.2	34.4	34.8	32.6	63.6	45.9
WISE	Llama3-8B	87.7	82.9	87.1	53.6	73.3	87.0	87.0	87.8	56.8	75.0
	Mistral-v0.3-7B	95.3	83.2	71.1	65.0	78.3	87.5	83.5	81.3	56.5	73.7
(b) True Success (%)											
IKE	Llama3-8B	99.4	99.6	99.3	82.6	92.7	99.5	99.1	99.1	78.1	90.8
	Mistral-v0.3-7B	88.7	79.6	72.1	56.7	72.5	94.1	89.6	91.3	65.7	81.7
ROME	Llama3-8B	29.6	36.8	20.7	9.2	21.2	20.9	23.0	24.3	11.6	17.9
	Mistral-v0.3-7B	11.1	3.6	3.2	6.1	7.0	10.4	3.5	10.4	22.9	14.5
FT-M	Llama3-8B	40.0	42.1	35.7	48.3	43.0	31.3	30.4	31.7	60.1	42.8
	Mistral-v0.3-7B	2.2	2.3	6.2	9.5	5.6	6.5	8.2	16.4	29.7	17.3
GRACE	Llama3-8B	3.1	6.8	7.5	17.5	10.0	18.4	27.8	22.6	31.7	25.6
	Mistral-v0.3-7B	0.6	0.4	1.8	17.9	7.6	5.6	6.5	6.1	33.8	17.0
WISE	Llama3-8B	9.6	17.1	15.4	10.4	11.7	19.8	27.0	26.5	14.5	19.5
	Mistral-v0.3-7B	4.3	3.6	6.1	9.9	6.7	5.2	2.6	7.0	9.4	6.8

Table 10: Per-question edit success across all 75 probes per instance, broken down by edit type, question category, and model. **CF** includes `ke_01`–`ke_02`, `cf_*`, and `zs_*`; **HF** includes `ke_03`–`ke_10` and `hb_*`. **All** is weighted by question count (25/10/10/30). *(a)* **Edit Success**: output contains `target_new`. *(b)* **True Success**: edit succeeds *and* output is free of generative degradation (`issue_type = ok`).

	`edit_success`				`issue_type`			
Method	n	raw	κ	α	n	raw	κ	α
IKE	60	0.983	0.000	0.000	59	0.966	0.000	−0.004
ROME	60	0.900	0.609	0.611	48	0.917	0.870	0.872
FT-M	60	0.833	0.674	0.668	23	0.696	0.233	0.199
GRACE	55	0.909	0.802	0.803	33	0.667	0.555	0.561
WISE	54	0.889	0.601	0.604	42	0.881	0.805	0.806
Pooled	289	0.903	0.737	0.737	205	0.859	0.780	0.780

Table 11: Per-method human–LLM agreement on `edit_success` and `issue_type`. `issue_type` is restricted to probes where both annotators marked `edit_success = 1`, after excluding probes with annotation errors on either side.